# Jiwei Zhou

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## EDUCATION

**Purdue University, West Lafayette, IN** **May 2023**
Ph.D. in Technology GPA: 3.98/4.0

**Shanghai Jiao Tong University, Shanghai, China** **June 2016**
M.Eng. in Software Engineering

**Zhuhai College of Jilin University (now Zhuhai College of Science and Technology), Guangdong, China** **July 2012**
B.E. in Mechanical Design & Manufacture and Automation

## JOURNAL PUBLICATIONS

J1. **Zhou, J.**, Camba, J. D., Company, P. (2026). CADialogue: A Multimodal LLM-Powered Conversational Assistant for Intuitive Parametric CAD Modeling. *Computer-Aided Design*, *191*, 104006. https://doi.org/10.1016/j.cad.2025.104006

J2. **Zhou, J.**, & Camba, J. D. (2025). The status, evolution, and future challenges of multimodal large language models (LLMs) in parametric CAD. *Expert Systems with Applications*, *282*, 127520. https://doi.org/10.1016/j.eswa.2025.127520

J3. **Zhou, J.**, Camba, J. D., Li, X. (2025). An Approach to Drawing Automation of Ship Stiffeners in the Shipbuilding Industry. *Computer-Aided Design and Applications*, *22*(1), 26-41. https://doi.org/10.14733/cadaps.2025.25-41

J4. **Zhou, J.**, & Hartman, N. W. (2024). A Model-Based Visual Inspection System (MBVIS) for Critical Plastic Bottle Dimensional Measurements. *Computer-Aided Design and Applications*, *21*(2), 270-280. https://doi.org/10.14733/cadaps.2024.270-280

J5. **Zhou, J.**, & Hartman, N. W. (2024). Development and Evaluation of a Vision Inspection System for Plastic Bottle Measurement. *Advances in Science and Technology*, *149*, 41-50. https://doi.org/10.4028/p-HPT9vc

J6. **Zhou, J.**, & Camba, J. D. (2021). Computer-aided process planning in immersive environments: A critical review. *Computers in Industry*, *133*, 103547. https://doi.org/10.1016/j.compind.2021.103547

## PEER-REVIEWED CONFERENCE PROCEEDINGS

C1. **Zhou, J.**, Camba, J. D., Company, P., Contero, M. *(accepted)*. EngDraw-VQA: An Agent-Based Framework for Automated Visual Question Answering Generation From Engineering Drawings. *Manufacturing Science and Engineering Conference (MSEC 2026)*.

C2. Company, P., Camba, J. D., Contero, M., **Zhou, J.** *(accepted)*. From conceptual to embodiment design using Sketch-Based Modeling and Feature Recognition Techniques. *23rd annual International CAD Conference (CAD'26)*.

C3. **Zhou, J.**, Camba, J. D., Company, P., Contero, M. *(accepted)*. Drawing-Checker: A Vision RAG Framework for Automated Comparison of Engineering Drawings. *36th CIRP Design Conference (CIRP Design 2026)*.

C4. **Zhou, J.**, Gupta, D., & Camba, J. D. Prompt2CAD: A Lightweight LLM Framework for Conversational CAD Generation and Iterative Refinement. *Procedia Computer Science*, *277*, 1663-1672. https://doi.org/10.1016/j.procs.2026.02.204

C5. Gupta, D., Camba, J. D., Fuerst, T., & **Zhou, J.** (2025). WIP: An AI-Based Virtual Assistant for Supporting a Large Engineering Course. *IEEE Frontiers in Education Conference (FIE 2025)*, https://doi.org/10.1109/FIE63693.2025.11328408

C6. **Zhou, J.**, & Hartman, N. W. (2023). A Framework for Model-Based Visual Inspection: A Case Study of Bottle Dimensional Measurements in the Plastics Industry. *2023 CAD Conference and Exposition*, 74-79. https://doi.org/10.14733/cadconfP.2023.74-79

C7. **Zhou, J.**, Camba, J. D., Hartman, N. W., & Li, Z. (2022). An Approach to Extend the Digital Thread From Requirements to Model Geometry. *Manufacturing Science and Engineering Conference (MSEC) 2022*. https://doi.org/10.1115/MSEC2022-80857

C8. **Zhou, J.**, Camba, J. D., & Fuerst, T. (2022). A Comparative Study on the Use and Interpretation of Annotated 3D Models. *IFIP 18th International Conference on Product Lifecycle Management*. https://doi.org/10.1007/978-3-030-94399-8_23

- 🏆 *Best Paper Award Nominee*

## MANUSCRIPTS UNDER REVIEW

J1. **Zhou, J.**, Camba, J. D., Company, P. CADThread: A Multimodal Engineering Design Dataset Linking Feature-Based Parametric CAD, Geometry, and Drawings. *Journal of Computing and Information Science in Engineering*.

J2. Camba, J. D., **Zhou, J.** GenAI-Aided Design in Engineering Education: Competencies, Challenges, and Opportunities. *Computer-Aided Design*.

J3. Gupta, D., Camba, J. D., Fuerst, T., **Zhou, J.** From Queries to Conversations: Exploring AI-Based Virtual Assistant Adoption, Usability, and Impact in Engineering Education. *ASEE Computers in Education*.

C1. **Zhou, J.**, & Camba, J. D. EngDraw-Extractor: A Multi-Agent System for Information Extraction from 2D Engineering Drawings. *International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE 2026)*.

C2. Camba, J. D., Contero, M., and **Zhou, J.** Large Language Models in Digital Manufacturing: An Empirical Study of Verification-Centered Design Cognition. *12th International Conference on Design Computing and Cognition (DCC'26)*.

## MANUSCRIPTS IN PREPARATION

C1. **Zhou, J.**, Camba, J.D. EngDraw-Annotator: Schema-Aware, LLM-Assisted Annotation for Engineering Drawings (Anticipated submission: 2026).

## CONFERENCE PRESENTATIONS

1. **ISM 2025 7th International Conference on Industry of the Future and Smart Manufacturing, University of Malta – Malta**: *Prompt2CAD: A Lightweight LLM Framework for Conversational CAD Generation and Iterative Refinement*
2. **2025 North American Plant Phenotyping Network (NAPPN) Annual Conference, Olivette, MO, USA**: *Integrating Artificial Intelligence in Plant Growth Monitoring for Innovative Rotary Aeroponic Systems*
3. **2023 10th International Conference on Mechanics, Materials and Manufacturing, Washington, D.C.:** *Development and Evaluation of a Vision Inspection System for Plastic Bottle Measurement*
4. **2023 20th Annual International CAD Conference (CAD'23), Mexico City**: *A Framework for Model-Based Visual Inspection: A Case Study of Bottle Dimensional Measurements in the Plastics Industry*
5. **2022 17th International Manufacturing Science and Engineering Conference, West Lafayette, IN, USA**: *An Approach to Extend the Digital Thread from Requirements to Model Geometry*
6. **2021 IFIP 18th International Conference on Product Lifecycle Management (Virtual Conference)**: *A Comparative Study on the Use and Interpretation of Annotated 3D Models*

## PROFESSIONAL EXPERIENCE

**Heliponix LLC, Evansville, IN** **June 2023 – Present**
**Computer Vision Scientist**

- Conduct interdisciplinary research in applied AI to drive innovation in engineering design, smart manufacturing, and technology education.
- Lead the deployment of AI-powered vision systems in IoT-enabled smart appliances, enabling real-time crop monitoring and autonomous decision-making.
- Collaborate across hardware and software teams to design cyber-physical prototypes, integrating imaging sensors, edge AI, and cloud-based analytics for scalable smart gardening and precision agriculture solutions.
- Develop and optimize machine learning and deep learning algorithms, influencing the design of computer vision hardware and software.

**Digital Enterprise Center, Purdue University, West Lafayette, IN** **August 2019 – May 2023**
**Graduate Research Assistant**

Drug Plastics - Model-based Metrology Project September 2021 – May 2023

- Developed an innovative and cost-effective Model-Based Visual Inspection System (MBVIS) to enhance the critical dimensional measurement accuracy of plastic bottles in pharmaceutical applications, utilizing Geometric Dimensioning and Tolerancing (GD&T) information extracted from a Model-Based Definition (MBD) dataset, demonstrating high agreement with a commercial visual inspection system through Bland-Altman plot analysis
- Utilized Computer Vision techniques (OpenCV-Python and convolutional neural networks) to develop a more efficient process to measure critical bottle parameters

WHIN - Digital Product Information Project August 2020 – August 2021

- Implemented RESTful APIs to facilitate interactive work instructions and Bill of Material (BOM) on a Product Data Management (PDM) system with Java
- Customized tooling management, process planning, workflow, project, and organizational structure on the PDM system to demonstrate to industrial partners

**Siemens Industry Software, Shanghai, China** **July 2012 – July 2019**
**Application Engineer**

- Led testing and development of CAD software (Siemens NX), improving product reliability and reducing customer-reported issues by 15%.
- Collaborated with cross-functional teams to design and automate test workflows, achieving >95% code coverage and improved development efficiency.
- Supported partner universities through Siemens GO PLM initiatives, delivering CAD training and fostering academic–industry collaboration.
- Mentored junior engineers and coordinated quality assurance processes across multiple international projects.

**Siemens Industry Software, Shanghai, China** **July 2011 – July 2012**
**NX Ship QA Intern**

- Tested on NX Ship Manufacture Module to improve ship product quality
- Created more than 300 reuse CAD parts to support NX Drawing Automation in the shipbuilding industry

## FUNDING & GRANTS

- **National Science Foundation (NSF) SBIR Phase II (Award No. 2151495)**—Partial support for research activities at Heliponix LLC
- **Project PID2022-137254OB-I00 (MCIN/AEI/FEDER, EU)**—Research support for peer-reviewed publications at Heliponix LLC
- **Wabash Heartland Innovation Network (Grant No. 4019008000 / 8000084103)**—Research support at Purdue University

## PUBLIC SPEAKING & OUTREACH

*Southwest Ecosystem Exchange*
**Evansville, IN** **June 2025**

- Represented the company at the inaugural Southwest Ecosystem Exchange, sharing our startup journey and key support resources leveraged—including SBIR/STTR, match funding, and local partnerships—to inspire and connect with regional innovation leaders.

*Generative AI and Programming-Based CAD*
**Purdue University, West Lafayette, IN** **Apr 2025**

- Delivered a 120-minute guest lecture covering programming-based CAD techniques, aligning with current research in Generative AI-enhanced design automation. Engaged graduate students through practical examples, interactive discussion, and Q&A.

*2025 CCO International Student Workshop – Internationally Friendly Employer Panel*
**Purdue University, West Lafayette, IN** **Mar 2025**

- Served as a panelist sharing insights on transitioning from international graduate studies to careers in the U.S. tech industry, emphasizing the role of AI and computer vision in practical applications.

*Leveraging Artificial Intelligence for a Competitive Edge in Business*
**Indiana University, Bloomington, IN** **Jan 2025**

- Delivered a keynote address on how businesses can leverage artificial intelligence for competitive advantage, highlighting opportunities, challenges, and strategies for adoption.

*Growing Smarter: How AI Helps Plants Thrive*
**University of Evansville, Evansville, IN** **Jan 2025**

- Delivered a presentation on the role of artificial intelligence in enhancing plant growth and agriculture to inspire the next generation of innovators of the UE Explorers program.

*Journey from Academia to Industry*
**Purdue University in Indianapolis, Indianapolis, IN** **Nov 2024**

- Presented a 30-minute talk to college and graduate students on transitioning from academia to industry, covering skill development, challenges, and career growth.

*Seeing with Smarts: How AI Helps Plants Grow!*
**Heliponix LLC, Evansville, IN** **Nov 2024**

- Conducted an interactive session introducing core concepts of artificial intelligence and computer vision to middle and high school students, engaging them with practical applications in industry.

## PROFESSIONAL SERVICE

**Technical Program Committee**

- 13th International Conference on Mechanics, Materials and Manufacturing (ICMMM), 2026

**Session Chair**

- 10th International Conference on Mechanics, Materials and Manufacturing (ICMMM), 2023

**Conference Reviewer**

- Institute of Industrial and Systems Engineers (IISE) Annual Conference & Expo, 2026
- International Manufacturing Science and Engineering Conference (MSEC), 2026
- American Society for Engineering Education (ASEE) Annual Conference & Exposition, 2023, 2024, 2026
- IEEE Frontiers in Education Conference (FIE), 2025, 2026

**Journal Reviewer**

- *Applied Soft Computing*
- *Artificial Intelligence and Applications*
- *ASEE Computers in Education*
- *Computer-Aided Design*
- *Computers & Graphics*
- *Computers & Industrial Engineering*
- *Design Studies*
- *Engineering Applications of Artificial Intelligence*
- *Engineering with Computers*
- *Expert Systems with Applications*
- *IEEE Transactions on Computational Social Systems*
- *Image and Vision Computing*
- *Results in Engineering*

- *SoftwareX*

## CERTIFICATION & TEACHING EXPERIENCE

Siemens Certified NX CAD Instructor **September 2011**

Siemens Teamcenter Instructor, PLM Product Development Center (PPDC), Shanghai, China **July 2013 – July 2019**

- Offered formal training in CAD skills and Teamcenter application to new employees

## AWARDS AND SCHOLARSHIPS

Conference Attendance Award **Feb 2025**

- NAPPN Conference 2025 Supported by USDA NIFA DSFAS Program

Purdue Graduate Research Assistantship **August 2019 – May 2023**

High-Performance Employee Awarded by Siemens PLM Software **2014 & 2016**

## LEADERSHIP EXPERIENCE

**Heliponix LLC, Evansville, IN** **June 2023 – Present**

***Research Associate Supervisor***

- Supervised a research associate, providing technical guidance and overseeing weekly progress on tasks related to plant science and computer vision

**Christian Students at Purdue (CSaP), Purdue University, West Lafayette, IN** **January 2020 – May 2023**

***Vice President***

- Supervised the business of the organization and organized a weekly Bible study for club members

**Siemens Industry Software, Shanghai, China** **July 2012 – July 2019**

***Mentor***

- Guided junior engineers and interns with professional CAD modeling and PLM application knowledge

## MEDIA COVERAGE

- *Southwest Ecosystem Exchange (2025).* Presentation on Anu's startup journey and ESO partnerships. [Link]
- *Purdue University Center for Career Opportunities & Office of Professional Practice (2024).* Invited panelist on international student career planning and transitions into U.S. industry roles. [Link]
- *Indiana University Kelley School of Business (2024).* Invited keynote speaker for MBA consulting project on Food as Medicine, presenting Anu's AI-driven indoor farming approach to health and sustainability. [Link]
- *Purdue University (2022).* Poster presentation on Realizing the Digital Enterprise projects, highlighted in university media coverage. [Link]